The Noisy Work of Uncertainty Visualisation

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Abstract

Uncertainty visualisation is quickly becoming a hot topic in information visualisation. Existing reviews in the field take the definition and purpose of an uncertainty visualisation to be self evident which results in a large amout of conflicting information. This conflict largely stems from a conflation between uncertainty visualisations designed for decision making and those designed to prevent false conclusions. We coin the term "signal suppression" to describe a visualisation that is designed for preventing false conclusions, as the approach demands that the signal (i.e. the collective take away of the estimates) is suppressed by the noise (i.e. the variance on those estimates). We argue that the current standards in visualisation suggest that uncertainty visualisations designed for decision making should not be considered uncertainty visualisations at all. Therefore, future work should focus on signal suppression. Effective signal suppression requires us to communicate the signal and the noise as a single "validity of signal" variable, and doing so proves to be difficult with current methods. We illustrate current approaches to uncertainty visualisation by showing how they would change the visual apprearance of a choropleth map. These maps allow us to see why some methods succeed at signal suppression, while others fall short. Evaluating

visualisations on how well they perform signal suppression also proves to be difficult, as it involves measuring the effect of noise, a variable we typically try to ignore. We suggest authors use qualitative studies or compare uncertainty visualisations to the relevant hypothesis tests.

1 Introduction

The phrase "uncertainty visualisation" can sometimes feel like a bit of an oxymoron. Uncertainty is often discussed as though it is an omnipresent spectre, touching every stage of our analysis. The fact that nobody really knows what it is, is either directly stated by authors who work in the field (Spiegelhalter 2017, Griethe & Schumann 2006) or implied by describing a myriad of things uncertainty could be without a concrete definition (Kinkeldey et al. 2014, Hullman 2016). Contrary to the murky waters of uncertainty, visualisations illuminates and unveils the unknowns in our data. This is illustrated with datasets such as Anscombe's quartet (Anscombe 1973) or the Datasaurus Dozen (Matejka & Fitzmaurice 2017, Locke & D'Agostino McGowan 2018) where visual statistics reveal elements of the data that are hidden to the typical summary statistics. This leads us to the conclusions that uncertainty visualisation is about revealing the unknown and shining a light on the more questionable parts of our analysis.

This sentiment is regularly echoed by the literature where we see authors express that visualising uncertainty is necessary for transparency and improved decision making (Padilla et al. 2022). Failing to incorporate the uncertain information in an analysis will lead to research being undermined as evidenced by a 6-month survey of anti-mask groups on Facebook during the COVID-19 pandemic that showed anti-maskers made persuasive arguments by exploiting inherrent uncertainty that was ignored by pro-maskers (Lee et al. 2021). Additionally, failing to present uncertain information can lead to poor estimation of failure probabilities (Der Kiureghian & Ditlevsen 2009), and poorer decision making both experimentally (Joslyn & LeClerc 2012, Savelli & Joslyn 2013, Kay et al. 2016, Fernandes et al. 2018) and in practice (Al-Kassab et al. 2014).

Given the importance of uncertainty visualisation, it would be nice if the field had some overarching guidelines, but this turns out to be easier asked for, than done. Rather than providing an overarching, cohesive narrative of the field, it is far more common to see a review paper lament the *difficulties* faced when trying to extract any cohesive insights. Kinkeldey et al. (2014) found most experimental methods to be ad hoc, with no commonly agreed upon methodology, formalisations, or greater goal of describing general principals. Hullman (2016) noticed there is a serious noise issue in the field, with noise from participants misunderstanding visualisations, misinterpreting questions, and incorrectly applying heuristics all overwhelming any information we can glean from studies. MacEachren et al. (2005) identified so much contradicting evidence that they were able to spend an entire page discussing the contradictions around a question as simple as "Should I map uncertainty to colour hue?". Griethe & Schumann (2006) did not identify common themes, but instead listed the findings and opinions of a collection of papers. Padilla et al. (2022) was able to synthesise some results cognative effects that arise in uncertainty visualisation experiments, however these effects were each discussed in isolation as a list of considerations an author might make rather than an overarching theory of rules for effective uncertainty visualisation. While these reviews are thorough in scope, none discuss exactly what we are trying to do hwen visualising uncertainty, or how the existing literature contribute to the broader goal of the field. The problem faced by the literature is easily summarised with a famous quote by Henri Poincaré.

> "Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house." - Henri Poincaré (1905)

That is to say, despite the wealth of reviews, the field of uncertainty visualisation remains a

heap of stones. This review attempts to address this issue by offering a novel perspective on the uncertainty visualisation problem. That is, we are going to use the wealth of established stone to construct the foundations on which we can build a house.

2 The purpose of uncertainty visualisation

Reading the uncertainty visualisation literature, we can become turned around as it often feels like we are reading several independent conversations at once. Largely because, we are.

Early mentions of uncertainty visualisation start to spring up in the late 1980s and early 1990s across several different fields (Ibrekk & Morgan 1987, MacEachren 1992), each with their own motivation for the work, with statistics often dappling in three. The work in computer science is motivated by issues in the publics perception of random variables, with the hope that visualisations would give laypeople the ability to extract important information from graphical representations (Ibrekk & Morgan 1987). This field views uncertainty visualisation as a human perception problem, where researchers often compare box plots, density plots, histograms, and other visual representations of a distribution to see which approach allows people to best extract means, probabilities, variances, etc. The methods discussed by psychology/economics also have similar foundations in the publics understanding of randomness, as they are concerned with using visualisations to help mitigate psychological bias that comes up when you try to communicate risk, a problem of concern since the early 1980s (Spiegelhalter 2017). These approaches often utilise pictographs and stress the importance of correct framing of the probabilities. Finally, around the early 1990s, discussions of uncertainty pop up in the cartography literature motivated by the inherent uncertainty in the measurements or estimates they are trying to visualise

within the constrained visual environment of maps (MacEachren 1992, Carr et al. 1992). This area views uncertainty visualisation as a problem in high dimensional visualisation, where uncertainty is just another variable that needs to be integrated into the visualisation (Griethe & Schumann 2006, Peña-Araya et al. 2025). As the field trudges on these distinct motivations bleed together and are now treated interchangeably as "uncertainty visualisation". We see computer science and psychology combine to form the research on "decision making" where researchers see which distribution representation elicits the best decisions and best mitigate psychological bias (Joslyn & LeClerc 2012, Savelli & Joslyn 2013, Kay et al. 2016, Fernandes et al. 2018). We see geoscience and computer science combine to evaluate map representations based on how well viewers can extract statistics about the distribution (Maceachren et al. 2012, Correll et al. 2018, Ndlovu et al. 2023).

This all begs the question, if theses fields do not agree on the purpose of uncertaitny visualisation, why have they intermixed so much that the literature treats them as interchangeable? There seems to be a single common thread that connects these approaches, that comes up time and time again across all subfields, and that is the belief that uncertainty visualisation is in pursuit of *transparency*. We see it comes up when authors claim that failing to include uncertainty is akin to fraud or lying (Hullman 2020, Manski 2020). We see it when authors assert that uncertainty communicates the legitimacy (or illegitimacy) of the conclusion drawn from visual inference (Correll & Gleicher 2014, Kale et al. 2018, Griethe & Schumann 2006). We see it when authors say uncertainty visualisations should communicate a degree of confidence (Boukhelifa et al. 2012) or validity (Hullman 2020, Griethe & Schumann 2006) in our conclusions. When authors suggest uncertainty visualisation should "guide, qualify, or soften our judgments of uncertain data" (Wilkinson 2005). When authors assert uncertainty visualisation should communivate a proportional

level of trust in our data that is more detailed than the binary results of a hypothesis test (Correll & Gleicher 2014, Correll et al. 2018). Combining these statements together, there is one glaring through line, and it is the assertion that uncertainty visualisation should act as some kind of visual hypothesis test where conclusions are visible when they are backed up by data, and invisible when they are not.

Therefore, the one fundamental motivator of the field is the idea that uncertainty visualisation should act as a "statistical hedge" for any inference we make using the graphic, and the other goals. Since the purpose of a visualisation is to give a quick gist of the information (Spiegelhalter 2017), this hedging should be communicated immediately through the graphic without the need for complicated mental calculations. Therefore, an effective uncertainty visualisation should not just "show" uncertainty, untrustworthy conclusions should *not be visible*. If we refer to the conclusion we draw from a graphic to be its signal and the uncertainty that makes this signal harder to identify as the "noise", we can summarise the above information into three key requirements. A good uncertainty visualisation needs to:

1) Reinforce justified signals to encourage confidence in results
2) Hide signals that are overwhelmed by noise to prevent unjustified conclusions
3) Perform tasks 1) and 2) in a way that is proportional to the level of confidence in those conclusions.

Usually visualisations that are unconcerned with uncertainty have no issue showing justified signals, but struggle with the display of unjustified signals. Therefore, we call this approach to uncertainty visualisation as "signal-suppression" since it primarily differentiates itself from from the normal "noiseless" visualisation approach through criteria (2). This is the main criteria we will use to assess uncertainty visualisations as we disentangle which approaches integrate uncertainty as signal, and which integrate uncertainty as noise.

3 Current Approaches

Even if transparency is the key pursuit of the field, most visualisations that are developed are developed with reference to their specific problem. That is, they are trying to come up with flexible distribution representations or new mappings that uncertainty could be mapped to in an existing graphic. These goals are not necessarily in conflict with the goal of transparency, so we will look at the existing approaches developed in the field and assess which successfully act as a statistical hedge, and discuss why, or why not, certain approaches do, or do not, work.

3.1 Ignoring uncertainty

The most common way uncertainty is integrated into a visualisation is actually to not do it at all. A survey conducted by Hullman (2020) found that very few authors actually include uncertainty in their visualisations, with only a quarter of respondents included uncertainty in 50% or more of their visualisations. Part of the abstinence of uncertainty visualisation comes from the fact that authors think that the plots will be overwhelming or they are uncertain how to calculate the uncertainty (Hullman 2020). This is not entirely unreasonable, given that even visualisation authors themselves seem to be in conflict about what exactly uncertainty is, and it is hard to assert people are ignoring it if we don't even know what it is they are ignoring.

3.1.1 What is uncertainty?

Describing what uncertainty actually is, is surprisingly hard. Most authors simply avoid the problem and describe the characteristics of uncertainty, of which there are plenty. Examples of these distinctions include whether or not the uncertainty is:

- Due to true randomness or a lack of knowledge (Begg et al. 2014, Spiegelhalter 2017, Gustafson & Rice 2019, Padilla et al. 2022, Hullman 2016, Walker et al. 2003)
- Quantifiable or unquantifiable (Spiegelhalter 2017, Walker et al. 2003, Padilla et al. 2022)
- Scientific or human (Benjamin & Budescu 2018, Gustafson & Rice 2019)
- Systematic or random (variance) (Sanyal et al. 2009)
- Statistical or bounded (Gschwandtner et al. 2016, Olston & Mackinlay 2002)
- In accuracy or precision (Griethe & Schumann 2006, Benjamin & Budescu 2018, Hullman 2016)

We could go on, there are enough taxonomies and typologies of uncertainty to fill several papers. It is the most common approach authors take to provide a definition, but the sheer volume of them makes it hard to see the method as anything other than a distinction without a difference, at least for our purposes. The distinctions are often accurate, but have little, if any, impact on how we should visualise, quantify, or discuss the uncertainty, bloating the literature and offering little in terms of genuine insights.

Rather than trying to define uncertainty by looking at the myriad ways in which it *does* appear in an analysis, we may find it easier to look at where it *does not*. Descriptive statistics describe our sample as it is and summarises large data down into an easy to swallow format. Descriptive statistics are not seen as the primary goal of modern statistics, however, this was not always the case. In 19th century England, *positivism* was the popular philosophical approach to science (positivists included famous statisticians such as Francis Galton and Karl Pearson). Practitioners of the approach believed statistics ended with descriptive statistics as science must be based on actual experience and observations (Otsuka 2023). In order to make statements about population statistics, future values, or new observations

we need to perform inference, which requires the assumption of the "uniformity of nature", that is, we need to assume that unobserved phenomena should be similar to observed phenomena (Otsuka 2023). Positivists believed referencing the unobservable was bad science. In other words, these scientists embraced descriptive statistics due to the inherent certainty that came with them. Since uncertainty is non-existent in descriptive statistics, it is clear that uncertainty is a by-product of inference. Looking at the previous classifications, we can see that they either offer information on what type of distribution to use (e.g. precision and bounded uncertainty are uniform distributions) or they offer suggestions in where to look to quantify the uncertainty (e.g. stage of the analysis, source of uncertainty, lack of knowledge, etc). These taxonomies likely describe what uncertainty is like or similar to because uncertainty is fundamentally statistical inference the cost of statistical inference.

The relationship between inference and uncertainty frequently appears in the literature and is one of the most consistent comments made, but it is often reffered to as a dependence on the "goals" of our analysis rather than inferential statistics. This has been known in some form or another since the earliest work on the subject, with Ibrekk & Morgan (1987) finding that "The performance of a display depends upon the information that a subject is trying to extract." Meng (2014) commented what is kept as data and what is tossed away is determined by the motivation of an analysis and what was previously noise can be shown to become signal depending on the question we seek to answer.

Otsuka (2023) suggested that the process of observing data to perform statistics is largely dependent on our goals, because the process of boiling real world entities down into probabilistic objects (or "probabilistic kind" as he puts it) depends on the relationship we seek to identify with our data. Wallsten et al. (1997) argue that the best method for evaluating or combining subjective probabilities depends on the uncertainty the decision maker wants

to represent and why it matters. Fischhoff & Davis (2014) looks at uncertainty visualisation for decision making decides that we should have different ways of communicating uncertainty based off what the user is supposed to do with it.

Spiegelhalter (2017) noted we "cannot assess the quality of risk communication unless the objectives are clear". Peña-Araya et al. (2025) asserted that uncertainty is context-dependent, and whether or not it is a source of doubt depends on the context. The communicating of this repeated point is clear, the role of signal and noise is not set in stone, it is flexible and can easily change. The only constant is that signal is our inferential statistic, and noise is the distribution of outcomes around that signal.

As a uncertainty is by product of statistical inference, it would make sense to simply quantify it as a distribution. Suggesting distributions as a representation of uncertainty is not new, Kay (2023) originally suggested thinking about uncertainty visualisations with distribution inputs, to replace the commonly used mean and standard deviation, removing the assumption of Gaussian distribution. Taking this point a step further we suggest that a distribution is sufficient to characterise *any* uncertainty, no matter how intangible it is. This is in contrast to the assertions made by some authors, who believe abstract notions of uncertainty, such as credibility (Thomson et al. 2005), forecaster confidence (Padilla et al. 2021), or uncertainty about uncertainty (Hadjimichael et al. 2024), are too complex to be quantified. This is simply untrue, as abstract concepts such as human belief or credibility are regularly quantified by Bayesians and hierarchical models are often used for uncertainty about uncertainty. Since uncertainty is always a by-product of statistical inference, it should always be possible to quantify it using a distribution.

3.1.2 Example: visualising estimates

If visualising uncertainty is fundamentally a scenario where we are visualising a set of random variables what does "ignoring" uncertainty look like? Figure 1 shows six different plots, a high and low uncertainty case for three different plots, all of which simply visualise the central mean of the distribution, which is what the uncertainty visualisation literature typically means when they say a method "ignores" uncertainty. Plots (a) and (b) show the line plot from a simple linear regression on the `mtcars` data, the regression in (a) has been done on the full dataset, and therefore has low uncertainty, while (b) was done on a sample of three points and has very high uncertainty. Plots (c) and (d) show a choropleth map, with county in Iowa being coloured by a simulated measured temperature, with plot (c) depicting a case where the measurement error is low, while plot (d) depictes a case where the measurement error is high. Finally, plots (e) and (f) depict a set of five groups of univariate variables we are looking to compare, where (e) depicts a case where the group variance is high (f) shows a case where the group variance is low. In all plots, we can only choose a single value to represent the distribution, so we opt for the average of all these distributions. In all plots, the variance in the plots in the top row is significantly different to the plots in the bottom row, so much so, that the trend should actually disappear. Can you see a difference between the plots in the top row versus the bottom row? Is the strength of the trend communicated through the visualisation? The answer to both of these questions is no as the high and low variance cases are identical.

These plots ignore uncertainty and as such, do not perform signal suppression. They don't even show a visual difference in the plot as the uncertainty changes. We will use these plots as a touchstone for our discussion on uncertainty visualisation, and return to them when we discuss the approaches suggested in the literature, assessing the approach on how well

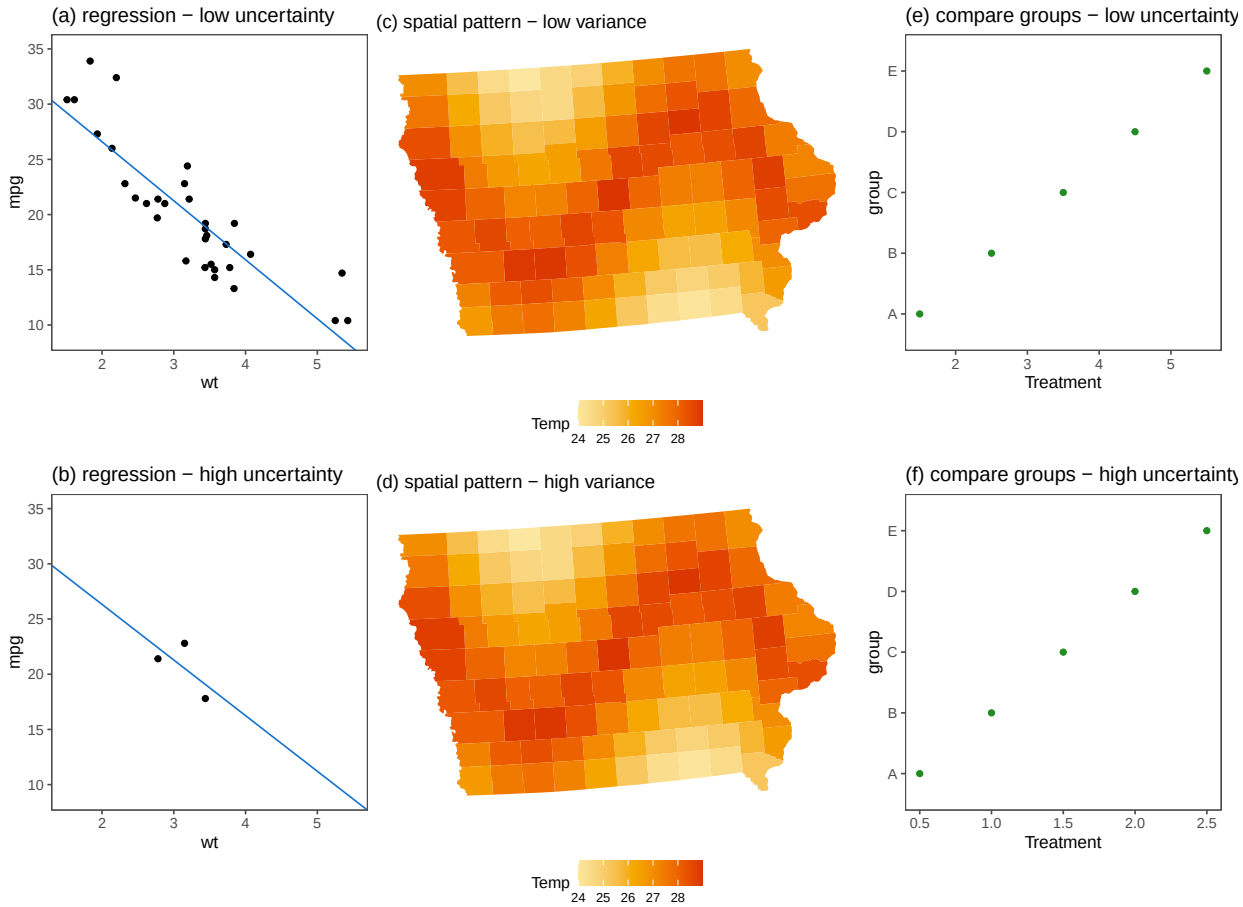

Figure 1: Six different plots, each depicting a high and low uncertainty case for three plots commonly seen in discussions around uncertainty visualisation. The plots are: (a) linear regression with low uncertainty, (b) linear regression with high uncertainty, (c) spatial trend with low uncertainty, (d) spatial trend with high uncertainty, (e) group comparison with low uncertainty, (f) group comparison with high uncertainty. The linear regressions have a downward slope, the spatial map has a sine wave pattern, and the groups show an increasing value in X. These patterns should be visible in the low uncertainty case, but almost invisible in the high uncertainty case. As each plot only shows the central estimate of each uncertainty value, the high and low scenario look identical for each plot.

it hides spurious relationships, and reveals the hidden variance structures in the data.

3.2 Uncertainty as a statistic

The most common approach to uncertainty visualisation (after just ignoring it all together) is to swap out our original statistic for one that shows uncertainty. The argument that an "uncertainty visualisation" should include visualisations of probability or variance is rooted in the decision making approach to the field, as research in this form is typically motivated by the psychological heuristics involved in interpreting uncertainty (Hullman et al. 2019). These days it pops up in all sub fields of visualisation. We see it as an exceedance probability map in geoscience (Kuhnert et al. 2018, Lucchesi et al. 2021), as an icon array in risk communication (Spiegelhalter 2017), or as a summary plot in statistics (Potter et al. 2010). Some authors take this approach because they explicitly believe uncertainty is a variable of importance in of itself (Blenkinsop et al. 2000), others strattle the line, asserting uncertainty is acting as signal and noise and should fulfil both roles (Peña-Araya et al. 2025). This seems a little too easy, can we just swap out the statistic for an "uncertainty statistic" and get an "uncertainty visualisation"? Is it that easy?

3.2.1 Example: visualising error

Figure 2 depicts the six visualisations from Figure 1, except the original central value estimate has been replaced with an uncertainty statistic. Visually these plots are identical with the visualisations remaining as a line plot, a choropleth map, and a scatter plot, with the change being in the meaning behind the plot rather than the visual appearance. Plots (a) and (b) are the most absurd examples of this visualisation approach, as replacing the estimated slope and intercept of a line with the standard error of said estimate produces

a plot that is inherently meaningless and is obviously completely unrelated to the data. This will not always be the case, as plots (c) and (d) are exceedance probability maps, real plots used in the literature to depict the probability of $P(X > 27)$ for the temperature of each county. Plots (e) and (f) simply show a scatter plot that depicts the variance of the groups instead of the estimate. The trend is no longer visible in the regression or the group comparison, but that is because the meaning of the plot has fundamentally changed, while the trend in the exceed probability map is still visible, although the meaning of this plot is also different.

3.2.2 What is an uncertainty visualisation?

When we refer to these graphics as "uncertainty visualisations" we are classifying visualisations by the data they display, not their visual features, even though graphics have never been defined by their input data. A scatter plot that compares mean and a scatter plot that compares variances are both scatter plots. Given that there is no special class of visualisation for *other* statistics (such the median or maximum) there is no reason to assume visualisations that simply depict a variance, error, or probability to be special.

The question about what is, or is not, an uncertainty visualisation seems to be one of the most pervasive divides in the literature. For example Wilkinson (2005) mentions that popular graphics, such as pie charts and bar charts omit uncertainty, and Wickham & Hofmann (2011) suggests their product plot framework, which includes histograms and bar charts, should be extended to include uncertainty. However, pie charts, bar charts and histograms have all been used as a significant number of experiments as examples of an "uncertainty visualisation" (Ibrekk & Morgan 1987, Olston & Mackinlay 2002, Zhao et al. 2023, Hofmann et al. 2012). What is going on here? Well, this problem seems

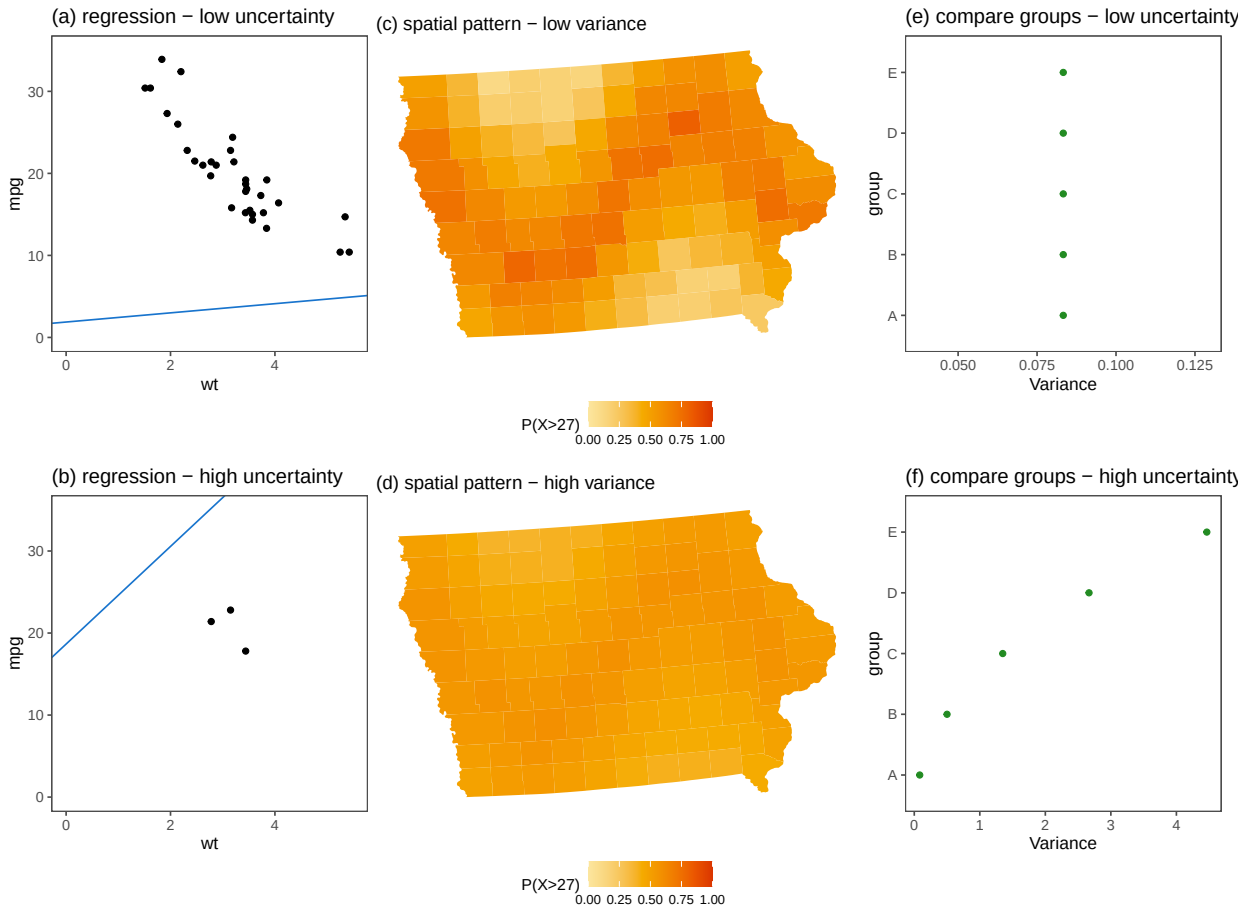

Figure 2: Six different plots, each depicting a high and low uncertainty case for three plots (regression, choropleth, and group comparison) commonly seen in discussions around uncertainty visualisation. Plots (a) and (b) have the standard error of the slope and intercept mapped to the slope and intercept of the line, plots (c) and (d) visualise the estimate value of P(X>27), and plots (e) and (f) visualise the mean of each group. We can see that there are some differences between the top and bottom row, but our visualisations are functionally meaningless in their new context. The standard error of a slope and intercept have no meaning when mapped to a line, the spatial pattern in the high and low spatial map is still visible. The group comparison is interpretable, but we are looking at a fundamnetally different visualisation.

to go back to the motivation of the field we discussed earlier. If you believe uncertainty visualisation is about communicating risks or random variables, uncertainty visualisations are just visualisations of "uncertainty statistics" or distributions. On the other hand you believe uncertainty visualisation is a dimensionality problem, then uncertainty visualisation is a function applied to an existing graphic that somehow adds the uncertainty in. The former has no limitation on the visual appearance of an "uncertainty visualisation", allowing pie charts, bar charts or histograms, so long as the graphic is visualising "uncertainty", while the latter believes uncertainty visualisations only exist in relation to some "normal" visualisation.

The fundamental problem with the statistic approach is that the uncertainty in the plot isn't acting as noise, it is acting as signal. This is because the "uncertainty statistic" has become the statistic we are drawing inference on, which mean our new statistic *also has uncertainty attached to it*, and therefore also have it's own uncertainty visualisation. Fundamentally, we have simply changed our inferential statistic, and in doing so, swapped the role of signal and noise. Of course, there is nothing wrong with explicitly visualising variance, error, bias, or any other statistic used to depict uncertainty as a signal. Just like any other statistic, these metrics provide important and useful information for analysis and decisions. However, this approach means that everything is an uncertainty visualisation, and if everything is an uncertainty visualisation, nothing is. Additionally, there is no interesting visual challenge associated with these graphics as they do not require any special visualisation techniques. The uncertainty in these graphics are acting as a signal variable, and they should be visualised using existing principles of graphic design

3.3 Uncertainty as a variable

If we need to include uncertainty in a plot, the enxt This approach treats uncertainty as a variable and is quite common, with many authors, across computer science (Kinkeldey et al. 2014), cartogophy (MacEachren et al. 2005), and statistical graphics (Wilkinson 2005) implicitly treating this "additional aesthetic" approach as the *only* method. We often see authors take this approach to uncertainty visualisation, not because we recognise the role in inference in our analysis, but rather because it is believed that displaying the features of one distribution across multiple plots makes the information hard to combine and results in some details (such as the estimate error) being completely ignored [Moritz et al. (2017); Correll2018].

3.3.1 Example: mapping independent varianbles

There are several approaches to mapping uncertainty to a spare axis in the plot. Kinkeldey et al. (2014) identified a split between intrinsic plots, where we map uncertainty to the colour or size of the geometric object of our estimate, and extrinsic plots where uncertainty is mapped to a separate geometric object glyphs or error bars. A similar distinction was made by Padilla et al. (2022) who classified uncertainty visualisations as graphical annotations (extrinsic), and probability mapped to a visual encoding channel (intrinsic), or a hybrid of the two. While these discussions to draw attention to how "integrated" the uncertainty is in the visualisation, it is unclear if this "intrinsic" plot

Figure 3 depicts the six visualisations Figure 1 except we have also mapped uncertainty to a spare aesthetic in the plot. The integrable approach is shown in plots (a) and (b) we have mapped the standard error of the slope to the width of the line, and the standard error of the intercept to the transparency. A bivariate map is shown in plots (c) and (d) and it

offers a slight divergence from the intrinsic/extrinsic approach as it visualises uncertainty within one variable, mapping both the estimate and it's variance to colour. The extrinsic approach is then shown in plots (e) and (f) map the estimate to a point, and map the variance to a separate geometric object, the length of a bar, to show a 95% confidence interval. We can see in all three maps, the trend is both visible in the top row where the variance is low, but also in the bottom row where the variance is high. While the high and low variance case do not look identical as they did in Figure 1, they communicate a message that says "here is the trend *and* here is the uncertainty" as two separate pieces of information, we are now visualising two independent signals, rather than a signal and it's noise. This method clearly is not visualising uncertainty as a "single integrated uncertain value" as Kinkeldey et al. (2014) would put it. The reality is, based on our discussion on inferential statistics, uncertainty *isn't* a separate variable, it is an component of the random variable that is indistinguishable from the random variable itself. It is worthwhile to interrogate why this occurs to see if we can move towards a version of this plot where we are able to communicate signal and noise as a single signal.

3.3.2 How to visualise a "single integrated uncertain value"

Typically, when making visualisations, we want the visual channels to be separable, that is, we don't want the data represented through one visual channel to interfere with the others (Smart & Szafir 2019). While this approach works for standard data visualisation, in uncertainty visualisation it allows the estimate and its variance to be read separately, preventing us from seeing the single integrated channel. This problem has actually popped up before in the literature as it has been repeatedly found that visualising uncertainty and signal separately causes the uncertainty to simply be ignored (Padilla et al. 2022). Additionally, there is research in visual semiotics on mapping uncertainty to intuitive aesthetic

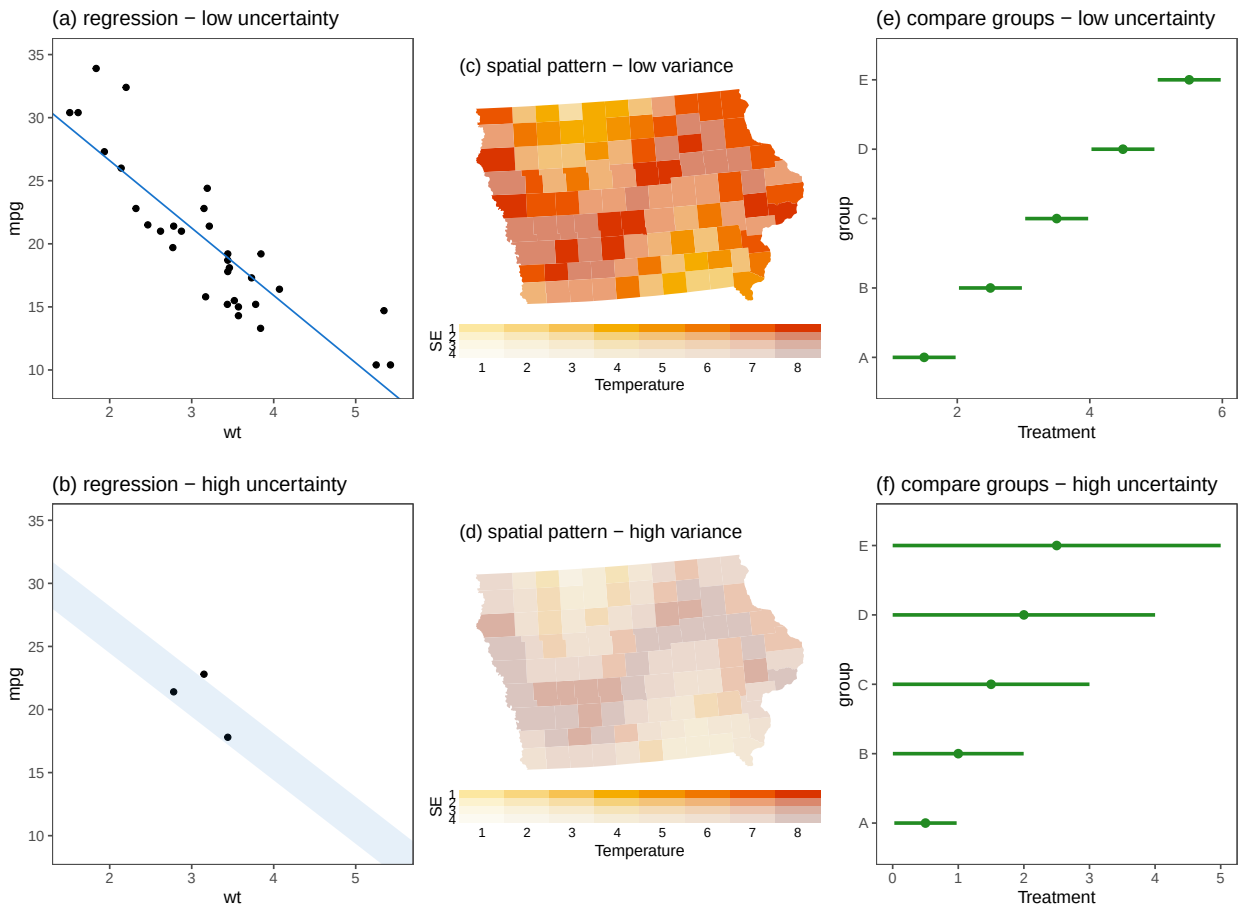

Figure 3: Six different plots, each depicting a high and low uncertainty case for three plots (regression, choropleth, and group comparison). Plots (a) and (b) have the standard error of the slope mapped to the line width, while the standard error of the intercept is mapped to the transparency. Plots (c) and (d) use a bivariate colour palette where the mean of the distribution is mapped to the colour hue, while the variance is mapped to the colour saturation. Plots (e) and (f) visualise the mean of each group as a point, and the uncertainty as an interval. We can see that there are some differences between the top and bottom row, but the 'signals' in the plots are still visible in the bottom row.

mappings, many of which interfere with the reading of the estimate such as mapping more uncertain values to lighter colours (Maceachren et al. 2012). To facilitate a visualisation of a single integrated value, maybe we should be leveraging integrability and visual semiotics to create a an ideal situation where the estimate and uncertainty are separately manipulable but read as a single channel by the human brain.

The problem is that even if we implement the most extreme versions of integrabiltiy our methods fall short. The bivariate map in Figure 3 already leveraged the fact that colour hue and brightness is integrable (Vanderplas et al. 2020) and decreasing saturation should make the colours harder to distinguish, but the signal is still clearly visible. However, the failure to perform signal suppression is not the only concern with relying on visual interference. The first problem is that the saturation relative to the signal is scaled arbitrarily with little concern for the strength of the signal. Another issue is that relying on perceptual interference to create statistical inference may not give us the level of control we need. Without a strong understanding of how these visual channels collapse down into a single channel, relying on integrability could create unintended consequences such as displaying phantom signals or hiding justified signals. Additionally, multi-dimensional colour palettes can make the graphics harder to read and hurt the accessibility of the plots (Vanderplas & Hofmann 2015). This is further complicated by the fact that human colour perception varies at an individual level. Some women are believed to have four different cone cells which allows them to perceive a greater range of colours, while others only have one or two types of cone cells and have colour deficiencies (Simunovic 2010). If this method were to work, we would need to calibrate each plot to an individual's ability to perceive colour. Instead of hoping that uncertainty might collapse signal values into a single dimension, we can do some of that work ourselves. As a matter of fact, some uncertainty visualisation

authors already have.

3.3.3 Another example: mapping combined variables

The Value Suppressing Uncertainty Palette (VSUP) (Correll et al. 2018), was designed with the intention of preventing high uncertainty values from being extracted from a map, blending colours together as they become more uncertain. Figure 4 depicts the Iowa choropleth map from Figure 3, but instead of the standard bivariate palette, we have used the VSUP that blends high variance colours together. Since the palette was designed with the extraction of individual values in mind and it has only been tested on simple value extraction tasks (Correll et al. 2018) or search tasks (Ndlovu et al. 2023), but we can see that, at least for our example, the high uncertainty spatial trend has functionally disappeared.

While the VSUP map does have signal suppressions, this colour blending approach creates a trade off visible signals in the plot. If we only use a single colour to express each random variable, we will need to decide which signals we suppress and which we do not. The VSUP in Figure 4 used the tree based method that was used by Correll et al. (2018), but Kay (2019) introduces several other methods and comments that which hypothesis are suppressed and which are not largely depends on the methods used in combining the colour palettes and the level at which the blending occurs. This means that the VSUP map is not suitable for exploratory data analysis, as we need to know the hypothesis we want to suppress before even generating the map.

3.3.4 Uncertainty and exploratory data analysis

We often think of visualisations as having two roles in our analysis. They can simultaneously be used for exploratory data analysis (EDA), which is the process of searching for interesting

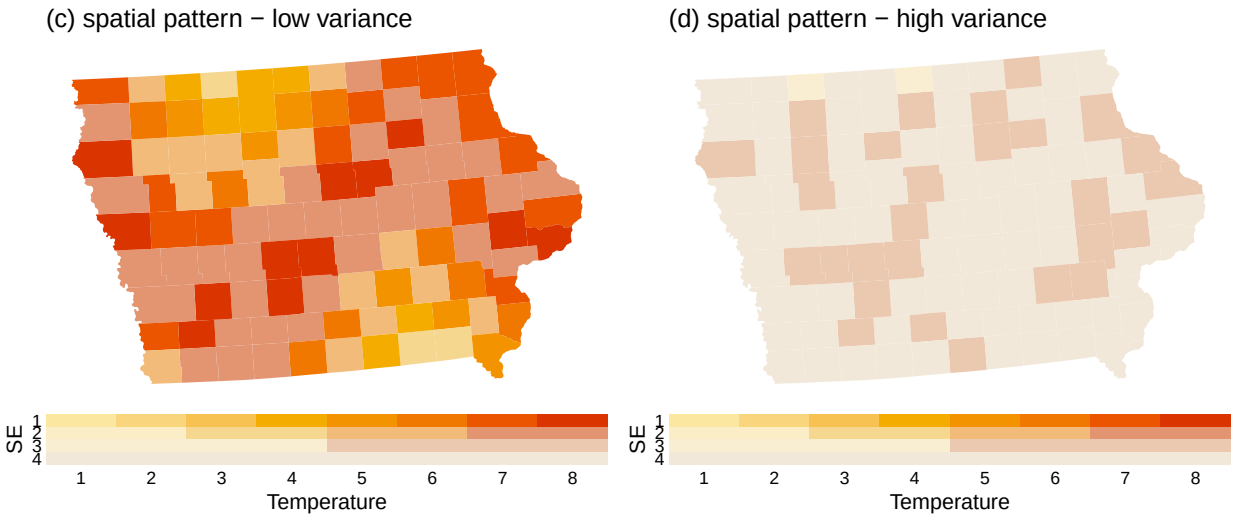

Figure 4: A variation of the choropleth map designed with a VSUP colour palette. The counties of Iowa are coloured acording to its average daily temperature and the variance in temperature. Similar to the bivariate map, temperature is mapped to hue while variance is mapped to saturation. Unlike the bivariance map, the colour space we are mapping our variables to has been transformed so that high variance estimates are harder to discern from each other. This map successfully reduces the visibility of the spatial trend in the high uncertainty case while maintaining the visibility of the spatial trend in the low uncertainty case.

hypotheses, and for confirmatory data analysis (CDA), the process of verifying a hypothesis (Tukey et al. 1977). Some have argued that the two methods are not entirely distinct, as it is often difficult to draw a hard line between EDA and CDA (Hullman & Gelman 2021). As with many concepts these approaches exist on a continuum, where we have an inherent trade-off between the number of hypotheses we can look for and the certainty of any conclusions reached. It can help to think of the knowledge-generating process of EDA and CDA as the nozzel on a hose with multiple spray options, where EDA is a fine misting spray that touches everything in the room, and CDA is a high-pressure jet capable of obliterating any and all debris from any single spot.

The EDA versus CDA dichotomy can be compared to the dichotomy between induction, for building theories, and deduction, for testing theories, from formal logic. One of the trade off's in the two methods is that conclusions deduction provide certainty, while conclusions from induction are inherently uncertain. This means that EDA, the inductive counterpart, has uncertainty in any conclusions reached through the method, with the requirement to follow up our new-found hypothesis with CDA if we want true certainty. This aligns with our earlier definition of uncertainty as descriptive statistics, one of the standard methods for EDA, have no inherent uncertainty as they do not look beyond the data.

This is why, despite the fact that an uncertainty visualisation system for EDA is one of the most discussed topics in the field (Hadjimichael et al. 2024, Peña-Araya et al. 2025, MacEachren et al. 2005, Sarma et al. 2024, Griethe & Schumann 2006), we have avoided the topic until now. Discussing EDA versus CDA gives us an additional layer of confusion, as authors frequently conflate the uncertainty in our data with the uncertainty in our conclusions that is inherent to the method. Authors often try to account for this inherent uncertainty and believe it should be incorporated into their plots, an argument

made explicitly by Hullman & Gelman (2021) when they said there is no such thing as a "model-free" visualisation. The fear of this inherent uncertainty leads authors to try to perform some kind of statistical super-test, where they pre-emptively hedge against every false inference that could possibly be drawn from a graphic (Guo et al. 2024, Savvides et al. 2019, Hullman & Gelman 2021). This is typically done by visualising a null distribution alongside the observed data on a single plot, an approach that is statistically identical to the line-up protocol (Buja et al. 2009). As a line-up protocols were designed to a graphical parallel to standard statistical hypothesis tests, a core tool of inductive statistics, they are deductive reasoning, and therefore not EDA. The sum total of a lot of CDA is not EDA, just as frantically swinging a high-pressure jet around a room is not equivalent to using a misting spray. Even if you do manage to wet everything, you are going to use an exorbitant amount of water to do it (and possibly also break a few things). This is not to say there is anything wrong with these approaches; they are incredibly useful tools for induction. The line-up protocol in particular is a useful alternative to hypothesis testing, as it allows for statistical inference when there is no classical test available or when tests involve multiple effects (Hofmann et al. 2012). Rather, the problem is that we cannot simultaneously perform EDA and inference as the approaches are, philosophically speaking, perpendicular to one another.

To truly create an uncertainty visualisation approach that is capable of EDA, we need to accept that uncertainty is inherent to the method, and it cannot be pre-emptively removed from the visualisations. If we accept this fact, then the only possible source of uncertainty in our "flexible uncertainty visualisation system" is from the data itself. That is to say, the only possible way for there to be uncertainty in a visualisation designed for EDA is that the data itself represents inference that has been done earlier in our analysis. We can see

this in the explanation of Figure 1 where the uncertainty in all cases, even measurement error, represents inference that was performed earlier in our analysis. This does not seem to be well understood in the uncertainty visualisation literature, where we see authors treat sampling and prediction distributions as though they are interchangeable (Hofman et al. 2020), despite the fact that they represent fundamentally different data.

Unlike in EDA, the signal suppression for VSUP is tied to our conclusions and not the data itself. Our ability to arbitrarily decide which values to blend at, or which suppression approach to use means that the "uncertainty" we are visualising will be informed by the conclusions we are drawing, and not a product of the data itself. Visualising the uncertainty in this way requires us to have already decided what signals we will suppress and which we will keep visible. This leads us to an additional, EDA based, requirement of our uncertainty visualisation system; the uncertainty and estimate need to be seen as a single unit, and any signal suppression should be informed by the data itself and should work the same regardless of any higher level hypothesis we might generate from the visualisation.

3.4 Uncertainty as a distribution

Instead of trying to summarise our random variables down into a single value or pair of values, why don't we just visualise the whole distribution? This is also a common sentiment in the literature, and we can see the approach in computer science with the HOPs (Hullman et al. 2015), in geoscience with pixel-maps (Lucchesi et al. 2021, Blenkinsop et al. 2000), and in statistics with spaghetti plots.

3.4.1 Example: visualising samples

Figure 3 depicts the six visualisations Figure 1 as distributions represented by either a sample of outcomes, or a quantile dot plot. Plots (a) and (b) show a linear regression as a sample of possible outcomes from the distribution. Plots (c) and (d) show a pixel map, which is a choropleth map with each county coloured by a sample of outcomes from the distribution. Plots (e) and (f) show the groups as quantile dot plots. We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the groups are all clearly visible in the low variance case, and are invisible in the high variance cases. In addition to successfully performing signal suppression, visualising the random variable as a distribution gives us additional information, such as the revealed bimodality of the distributions in (e) and (f). This means the graphic also achieves the third criteria for signal-suppression, i.e. our difficulty in seeing the distribution is proportional to the level of uncertainty in the graphic.

3.4.2 Quantified versus unquantified uncertainty

The visualisation in Figure 5 performs signal suppression, without sacrificing the viewers ability to extract general statistics, unless those statistics *should* be harder to extract due to the uncertainty in the value. By showing our distribution as "data" we are able to read the "uncertainty" plots using the same perceptual tasks that are used to read the "non-uncertainty" plot. This is also easier on the reader, as Ndlovu et al. (2023) found that people read complicated bivariate and VSUP plots the same way they read choropleth maps, even it is invalid. This approach also does not significantly hinder our ability to extract the individual values mapped by the previous plots, as extracting global statistics from a sample can be done with relative ease (Franconeri 2021). The success of this approach

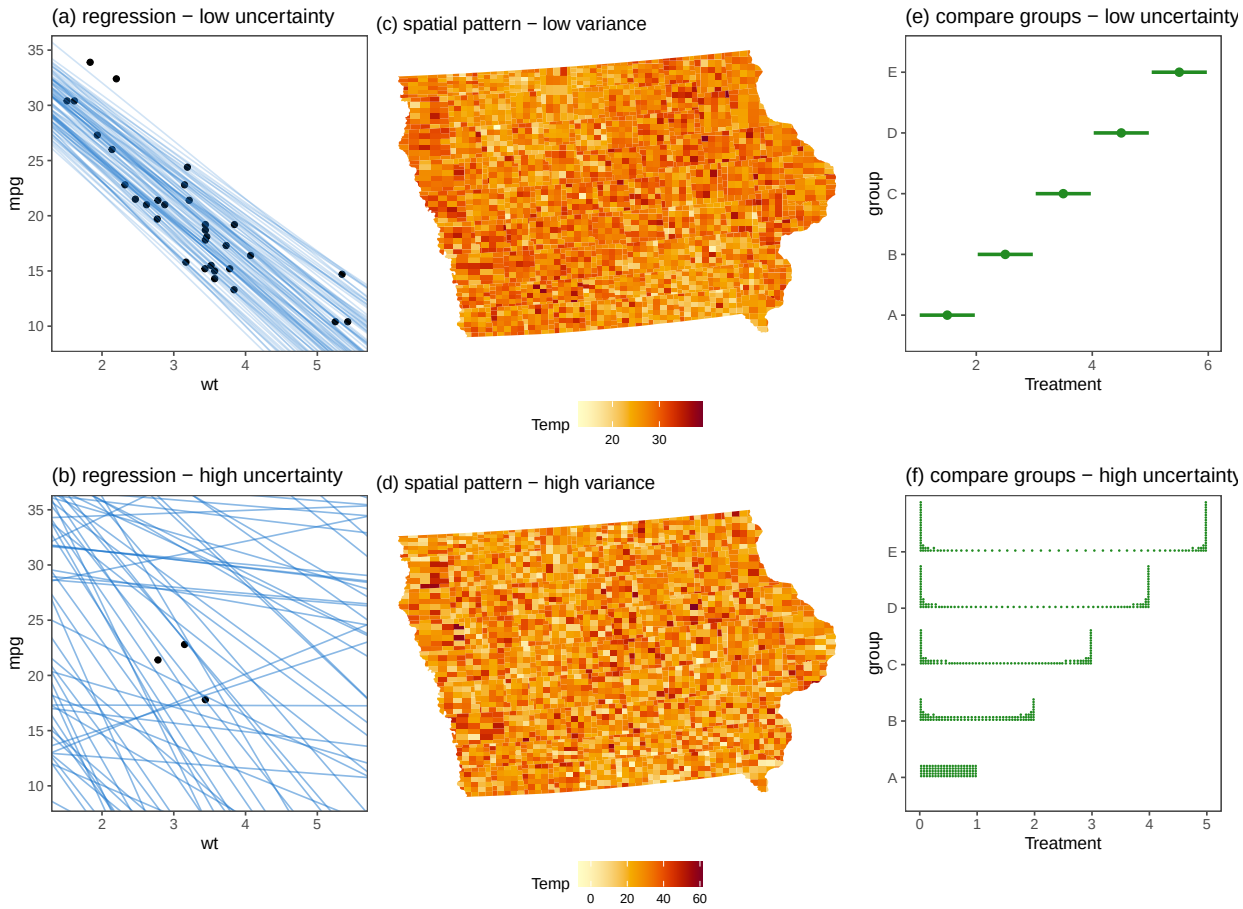

Figure 5: Six different plots, each depicting a high and low uncertainty case for our three examples: regression, choropleth, and group comparison. Plots (a) and (b) show a linear regression as a sample of possible outcomes from the distribution. Plots (c) and (d) show a pixel map, which is a choropleth map with each county coloured by a sample of outcomes from the distribution. Plots (e) and (f) show the groups as quantile dot plots. We can see that the strong downward trend in the linear regression, the sine wave in the choropleth map, and the incremental increase in the groups have all disappeared.

highlights an interesting problem for uncertainty visualisation. We are able to include more information by offloading more computation to visual processing, but should we offload *all* computations to the visual processing and just visualise the raw data?

This is, in a way, a question about how much of our uncertainty should be quantified. Quantified uncertainty usually focus narrowly on concepts such as probability, confidence intervals, variance, error, or precision (Hullman et al. 2018, Maceachren et al. 2012, Thomson et al. 2005), while unquantified uncertainty often includes a broader range of concepts like missing values, reliability, model validity, or source integrity (Griethe & Schumann 2006, Wilkinson 2005, Pang et al. 1997, Pham et al. 2009, Boukhelifa et al. 2017). When discussing uncertainty we typically include these unquantified uncertainties, not because these things *are* uncertainty, but because they can *create* uncertainty when we perform inference by violating the assumption of the uniformity of nature.

If the assumption violations are relevant, and we have the tools to visualise them, we should. For example, in the event of missing data, we can use the `naniar` package Tierney & Cook (2023) to include them as a "shadow" in our visualisations to check if missing values have structure. This approach amounts to just "showing the data" which is a simple but effective option for uncertainty visualisation that is largely overlooked. While this approach is easy (when the tools for it exist) this approach will not eliminate trends that have become invalid due to structure in missing data or an invalid model. We can only integrate uncertainty as noise when that uncertainty has been *quantified* as an effect on the estimates we are visualising. This is not to say you should do one method over another, visualising both quantified and unquantified uncertainty is necessary for a healthy analysis. Data analysis often works in cycles, where we find assumption violations using EDA, quantifying the effect of these violations on our inference, and then visualising the output of that inference using

uncertainty visualisation to perform even more EDA. Rather, our point is that uncertainty is a living breathing component of our visualisations that is always changing, and we should shine a light on it's shape with visualisation whenever we get the chance.

4 Evaluating uncertainty visualisations

If we want to make conclusions about how effective any uncertainty visualisation method is we need to look at the results of evaluation experiments. Unfortunately, despite the abundance of these studies, existing literature reviews have struggled to synthesise them into any common rules (MacEachren et al. 2005, Kinkeldey et al. 2014, Hullman 2016). This issue is particular to uncertainty visualisation, and pervasive, so it begs the question, what is causing it?

There are as many explanations for the noisy evaluation studies, as there are contradictions in the research itself. Kim et al. (2019) believes there is some interference from participants prior beliefs; Hullman (2016) believes the noise in the literature could come from visual heuristics, subjective probabilities, unknown participant utility functions, or misunderstanding of statistical concepts (such as confidence intervals). Kinkeldey et al. (2014) suggest the best visualisation changes depending on the audience, so we cannot expect the same results between different subpopulations. Brennen & Tuerk (2018) attributes issues to cognitive load from complicated uncertainty visualisations, and the participants prior experience in a topic, suggesting it is aleviated by measuring confidence and cognative load. While these issues will certainly have some impact on our ability to create a cohesive theory, most of them are not unique to uncertainty, visualisation and fail to explain the problems in this field in particular. Rather, the issue is likely coming from a disconnection between the evaluation method and wider goal of the field.

4.1 Current evaluation methods

The inconsistency in the motivations for uncertainty visualisation bleed into the evaluation of the methods. Treating the results of these papers, as though they are somehow agnostic towards their stated purpose, leads to regularly conflicting results in the field. This problem is pervasive within standard visualisation evaluation literature, where studies are regularly criticised for failing to correctly align questions and plot types (Vanderplas et al. 2020) as though the goals of our analysis and the plots we use are somehow independent. This becomes particularly egregious in uncertainty visualisation because, unlike other statistics, uncertainty plays two, opposite, roles in our analysis. The role of signal is to be visible, while the role of noise is to obfuscate, so flipping the role of uncertainty also flips our conclusions about a methods effectiveness. This problem was also noticed by Brennen & Tuerk (2018), however their solution of measuring cognitive load and confidence would not alleviate this deeper problem. This issue is further aggravated by the fact that authors often provide little justification for their chosen response models (Hullman 2016), and the fact that visualisations are compared on the basis of being "uncertainty visualisations" (Ibrekk & Morgan 1987, Hullman et al. 2015, Hofman et al. 2020), a classification that holds little to no meaning. This disregard for the wider motivation of the field leads to findings that are confusing and conflicting, which cannot be coalesced into a cohesive theory.

Uncertainty visualisations are most commonly evaluated based on how accurately viewers can extract an estimate and its variance (Kinkeldey et al. 2014, Hullman et al. 2019). This is not unusual, as direct observation is the simplest way to verify that information can be accurately read from a graph (Vanderplas et al. 2020). Unfortunately, this approach doesn't work for uncertainty visualisation. The second we ask a specific question about a statistic, that statistic becomes our inferential statistic, even if the plot was not the in-

tent behind the plot. By shifting the focus from $\hat{X}$ to $Var(\hat{X})$ or $P(\hat{X} < x)$ we end up evaluating visualisations on their ability to convey uncertainty statistics, rather than their ability to perform signal suppression. There is nothing unique to uncertainty in these studies, so when we boil the findings down to generalised results, they simply restate existing principles within information visualisation, even if they do not realise it themselves. Some of the findings are particularly obvious, such studies that found participants were more accurate when reading a probability expressed as a text than when they had to extract if from a graphic (Cheong et al. 2016, Savelli & Joslyn 2013). Other studies replicate existing research, such as the finding that a probability mapped to a position is more accurate than one mapped to area (Ibrekk & Morgan 1987, Gschwandtner et al. 2016), a finding that follows the hierarchy of perceptual tasks, which has been established for over 40 years (Cleveland & McGill 1984). This extends beyond simple accuracy evaluations, as we found that colour was more effective than size when searching for maximums and minimums in variances (Sanyal et al. 2009), even though we have known pre-attentive aesthetics such as colour are more efficient for search tasks since the 1980s (Vanderplas et al. 2020). By classifying these studies as evaluations of "uncertainty visualisation" while evaluating uncertainty as signal, we are encouraged to see successful examples of signal-suppression as failure. This approach leads authors advise against particular aesthetic mappings for uncertainty, because it leads to participants have more difficulty extracting values (Blenkinsop et al. 2000). This conclusion is antithetical to the goals of signal-suppression and occurs because these methods evaluate uncertainty as a signal, not as noise.

If we cannot directly measure uncertainty for fear that it turns into signal, we might then assume we can measure the secondary benefits of increased transparency. This seems to be the approach of many visualisation authors, as secondary benefits such as trust, confi-

dence, and risk aversion are all frequently used in uncertainty evaluation studies (Hullman et al. 2019). Unfortunately, measuring these secondary effects often leads to confusing conclusions that simultaneously argue for, and against, the inclusion of uncertainty. This is most commonly noticed in the use of trust as a measure, as several authors have commented that measuring trust, and not transparency, can lead to a questionable subtext that argues against transparency (Spiegelhalter 2017, O'Neill 2018). Confidence also has the confounding factor that there are many reasons for decreased confidence, including a poorly constructed visualisation, and using it as an evaluation metric collapses these sources down into a single metric. This subtext pops up in the uncertainty visualisation literature where visualisation authors will opt to leave out uncertainty information because it decreases confidence in the conclusions (Hullman 2020). A sentiment that carries over into evaluations of uncertainty visualisation, where Blenkinsop et al. (2000) commented that visually integrable depictions of uncertainty should be avoided, as they decrease the viewers confidence in their extracted data values. Another secondary effect that is similar to trust and confidence is risk aversion, which is an economics term used to describe an agent who would chose a random variable with a lower expected payout because it also has a lower variance. Risk aversion's role in the uncertainty visualisation is, to be frank, confusing, as it is unclear whether we shoud want more rational decisions that ignore uncertainty, or less rational decisions that incorporate uncertainty. The uncertainty visualisation literature does not clear this up as authors will argue that more risk aversion is better in one paper (Hullman et al. 2019), and argue for less risk aversion (by proxy of suggesting rational agents as a benchmark) in the next (Wu et al. 2023). Ultimately, these approaches have similar issues to the value extraction studies, except they are slightly more confusing in their goals leading them to simultaneously argue for an against the inclusion of uncertainty

in a visualisation.

This is made even more complicated by the fact that there is no standard technique to evaluate uncertainty as noise, and attempts to do so are often ad hoc with varying levels of success. Some authors try to incorporate including uncertainty by asking vague questions but compare the results to a ground truth, creating a situation where plots are *evaluated* as a value extraction study without participants being asked the *questions* of a value extraction task. Examples include asking participants for the "best estimate" (Ibrekk & Morgan 1987), or identify when a distribution is "furthest to the right" (Hofmann et al. 2012) and then compare these results to the distributions expected value. The convoluted questions designed to incorporate uncertainty information not only lead to noisy results, but frequently confuse the participants and lead to obviously wrong responses. For example, Hullman et al. (2015) asked participants to compare two groups, A and B, and had many participants say that group A was more likely to be bigger, despite group B having a higher mean. In another study Gschwandtner et al. (2016) ask participants the "probability that the interval has already ended at the marked point in time?" and participants replied with the probability that the interval had already started. The confusion from trying to incorporate uncertainty uncertainty can extend to the authors of a study, who often tie themselves in knots trying to come up with measurements for the effects of noise. We can see an example of this in the two Padilla et al. (2017) where the first study required participants to assume that an oil rig being "more likely to be hit" by a hurricane would *not* translate to the rig sustaining "more damage", while the second required participants to assume the opposite. This clearly contradictory assumption in how to appropriately use uncertainty information seemed to go unnoticed by the authors. While these approaches are certainly a step in the right direction, the experiments overcomplicate the problem and

produce results that are not even consistent within the individual experiment.

This is not to say all all evaluation studies fail to properly evaluate uncertainty as noise. There are several studies that ask participants to identify a particular signal that the noise is trying to obfuscate (Kale et al. 2018, Correll & Gleicher 2014) which seems to be an effective method. The only problem with these studies is that most uncertainty visualisation methods are not integrated into *the grammar of graphics* (Wilkinson 2005, Wickham 2010) so we regularly see comparisons between disparate plots that that would never be considered substitutes for one another outside the artificial "uncertainty visualisation" framework they are placed within. For example Kale et al. (2018) compared static bar charts with error bars to a bar chart with animated samples, meaning that any difference in participants ability to read the plot could be due to the statistic (confidence interval versus sample), the geometry (bars versus intervals), or the use of animation (static versuss animated plots). This means that even when evaluations *are* done correctly, there is no generalisable theory we can take from the results.

The point here is not to mock the existing literature for poor academic merit. If you were to go read these papers yourself, you would find them to (usually) be logically consistent and well formulated pieces of work. Rather, the point is to illustrate that evaluating uncertainty as a noise is surprisingly difficult. Designing tests for signal suppression will require a formalisation of uncertainty within the grammar of graphics, as well as improved evaluation methods.

4.2 Implicit Hypotehsis Testing

The main problems with current uncertainty visualisation evaluations is that they often require explicit (or convoluted) questions about the variance. Asking direct questions

about the statistics or outcomes is not an explicit requirement of visualisation evaluations. In their review of testing statistical graphics, Vanderplas et al. (2020) drew a distinction between explicit tests, where participants are asked direct questions about specific features of a plot, and implicit testing, where users identify both the purpose and function of the plot. The lineup protocol is the most salient example of the implicit approach. Lineups are a confirmatory visualisation tool where participants are shown a set of $M - 1$ plots, and asked to identify the plot that is the "most different", leaving participants to decide what "most different" means to them, even if it is not what the authors intended (VanderPlas & Hofmann n.d.). The implicit testing does not limit the versaility of the approach, with the lineup being used to evaluate the effectiveness of different types of plots (Hofmann et al. 2012), colour palettes (Reda & Szafir 2021), and design decisions (VanderPlas & Hofmann n.d.).

Lineup protocols are not only a useful for their implicit testing, but they also have parallels to hypothesis testing that can be leveraged for uncertainty visualisation. The concept of signal suppression is, at its core, an assertion of statistical validity, which means the visibility of signals should be directly proportional to p-values or some equivalent measure. This comparison is not new in uncertainty visualisation where parallels have been drawn to frequentist statistics by Correll & Gleicher (2014) who compared results to Cohen's D, and to Bayesian statistics by Kim et al. (2019) who evaluated plots based on their impact on users prior beliefs. The comparison to hypothesis testing is far more natural for the lineup protocol, with Majumder et al. (2013) explicitly defines a visual test statistic, while Li et al. (2024) compared standard statistical tests to the lineup-protocol using power curves. The parallels to line-ups and uncertainty visualisation are numerous, with the relationship even being identified by Hullman et al. (2015) in their development of hypothetical outcome

plots.

The lineup protocol and uncertainty visualisation share a lot in common, with the motivation behind lineups also rooted in checking if perceived patterns are real or merely the result of chance (Buja et al. 2009, Wickham et al. 2010). As both approaches are attempting to do the same thing, it is likely that we are unable to leverage the lineup protocol directly to evaluate uncertainty visualisation, but new evaluation methodology should try to learn from the success of the lineup approach. Designing an implicit testing method for uncertainty visualisation that allows us to draw parallels to standard notions of statistical significance, would solve many of the issues with the current evaluation approaches.

4.3 Heuristics research

There is one element of uncertainty visualisation research that we have not touched on yet, but is incredibly consistent, and that is the research around visual heuristics. The uncertainty visualisation heuristics research, unlike the plot evaluation work, is increbidly consistent, likely because heuristics are indifferent to the goal of our analysis and uncertainty's role as signal or noise. Several heuristics are of particular importance for uncertainty visualisation. We will give a brief overview of them here, and for a more in depth analysis, we recommend reading Padilla et al. (2022) as they provide a more in depth review.

The use of confidence intervals or boundaries have a few known issues in their interpretation. The sine illusion can cause the confidence interval of a smoothed sine curve to seem wider at the peaks than the troughs, causing us to underestimate uncertainty associated with changing values (Vanderplas & Hofmann 2015). Similarly researchers frequently find that visual boundaries such as the cone of uncertainty in hurricane forcasts (Padilla et al. 2022), or confidence bands or ellipses (VanderPlas & Hofmann n.d.) create the impression of a

meaningful bound, separating points outside the bounds from those within it which can create problems if the bound is not inherently meaningful.

While many researchers tend to prefer frequency framing, they are not without their critiques. Points that were on an outcome of an ensemble display were perceived as more likely than points not on an outcome, even when the point that was not on an outcome was closer to the centre of the distribution (and therefore more likely) (Padilla et al. 2017). This is considered to be a variation of the within bar bias, where outcomes within the bar on a bar chart as percieved to be more likely than those outside it (Newman & Scholl 2012).

We should also try and avoid visualsing central estimates if possible as several studies have found that viewers ignore uncertainty and compare the difference between the two means if they have the option (Padilla et al. 2022).

These heuristics have the potential to create noise in signal suppression evaluations, and should be kept in mind when designing uncertainty visualisations as well as evaluation experiments.

5 Conclusions and Future work

This paper has identified gaps in the uncertainty visualisation literature that must be filled for the field to progress. We formalised the uncertainty visualisation problem, and in doing so highlighted an untouched area of uncertainty visualisation research.

Experimental practices on uncertainty visualisation need to be standardised. Uncertainty visualisations for decision making treat uncertainty as signal, while visualisations for signal-suppression treat uncertainty as noise. As the literature currently exists, there is no way to combine papers to get a meaningful sense of how uncertainty information is understood

by a viewer. Researchers need to ensure that when they identify the motivation behind their visualisation technique, and that their evaluation methods match the motivations of the paper. Additionally *experimental methods that evaluate uncertainty as noise need to be developed.*

Experimenters should consider evaluating visual aesthetics on how well they suppress information in a graphic. Research into separability and integrability is of particular interest to uncertainty visualisation, as it allows interference from the uncertainty variable. When designing experiments, authors often choose aesthetics that are visually distinguishable; uncertainty visualisation authors should consider doing the opposite.

Software that allows users to easily perform signal suppression should be developed. Existing uncertainty visualisation methods view a distribution as its own object and there are no software options for the "an uncertainty visualisation is a function of an existing visualisation" philosophy.

Signal suppression is an untouched area of visualisation research and developing methods for the practice may require us to challenge our entire notion of what makes a good visualisation. We are interested to see how this challenge plays out.

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et al. 2009), and `rgeos` (Bivand & Rundel 2023). The GitHub repository for this paper can be found at https://github.com/harriet-mason/ARSA-UncertaintyLitReview which contains the files required to reproduce this article in full.

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